

COMS3003A Test 2

This test contributes 10% to your final mark

Date: 13/03/2026

1. **(4 Points)** For the Turing Machine described in Table 1, answer the following questions:
 1. **(2 Points)** Provide the time function. $t(n) = O(n^2)$
 2. **(2 Points)** Provide the space function. $s(n) = O(n)$
2. **(6 Points)** Prove (or disprove) that the following configurations are a computation of the Turing Machine described in Table 1
 1. **(3 Points)** $q_{\text{init}}011, q_111, 1q_11, 11q_1 \sqcup, 1q_21, q_31, q_7 \sqcup, q_{\text{init}} \sqcup, q_{\text{accept}} \sqcup$
 2. **(3 Points)** $q_4111, 1q_411, 11q_41, 111q_4 \sqcup, 11q_51, 11 \sqcup q_{\text{reject}} \sqcup$

Let $c_i \vdash^{I_k} c_{i+1}$ mean that $c_i \vdash c_{i+1}$ by instruction $I_k \in \delta$

Let $c_i \not\vdash c_{i+1}$ mean that there is no $I \in \delta$ such that $c_i \vdash^I c_{i+1}$

1. $q_{\text{init}} \xrightarrow{I_2} 011 \vdash q_1 \xrightarrow{I_6} 11 \vdash 1q_1 \xrightarrow{I_6} 11q_1 \sqcup \vdash 1q_2 \xrightarrow{I_4} 1q_3 \xrightarrow{I_9} 1q_7 \xrightarrow{I_{12}} q_7 \sqcup \vdash q_{\text{init}} \xrightarrow{I_{22}} \sqcup \vdash q_{\text{accept}} \xrightarrow{I_1} \sqcup$
 - Every configuration pair is yielded by some instruction and the previous configuration (+1 Mark if they have shown this like above or in some other equal method)
 - $q_{\text{init}}011$ is the initial config for $C(M, 011)$
 - The last configuration is the only configuration in a halting state (+1 Mark if **BOTH** halting and initial config specified | No half marks)
 - Therefore the sequence is the computation $C(M, 011)$ (+1 Mark for correct conclusion iff it follows correctly)
2. $q_4 \xrightarrow{I_{15}} 111 \vdash 1q_4 \xrightarrow{I_{15}} 11q_4 \vdash 11q_4 \xrightarrow{I_{15}} 111q_4 \sqcup \vdash 11q_5 \xrightarrow{I_{13}} 11q_5 \vdash 11 \xrightarrow{I_{18}} \sqcup q_{\text{reject}} \sqcup$
 - Every configuration pair is yielded by some instruction and the previous configuration
 - The last configuration is the only configuration in a halting state
 - q_4111 is not the initial config of $C(M, 111)$ as q_4 is not the initial state
 - (+1 Mark “Not initial config”)
 - (+1 Mark for why it is not the initial config)
 - Therefore the sequence is *not* the computation $C(M, 111)$ (+1 Mark for correct conclusion iff it follows correctly)

I_1	$q_{\text{init}}, \sqcup \rightarrow q_{\text{accept}}, \sqcup, R$	I_{13}	$q_4, \sqcup \rightarrow q_5, \sqcup, L$
I_2	$q_{\text{init}}, 0 \rightarrow q_1, \sqcup, R$	I_{14}	$q_4, 0 \rightarrow q_4, 0, R$
I_3	$q_{\text{init}}, 1 \rightarrow q_4, \sqcup, R$	I_{15}	$q_4, 1 \rightarrow q_4, 1, R$
I_4	$q_1, \sqcup \rightarrow q_2, \sqcup, L$	I_{16}	$q_5, \sqcup \rightarrow q_{\text{reject}}, \sqcup, R$
I_5	$q_1, 0 \rightarrow q_1, 0, R$	I_{17}	$q_5, 0 \rightarrow q_6, \sqcup, L$
I_6	$q_1, 1 \rightarrow q_1, 1, R$	I_{18}	$q_5, 1 \rightarrow q_{\text{reject}}, \sqcup, R$
I_7	$q_2, \sqcup \rightarrow q_{\text{reject}}, \sqcup, R$	I_{19}	$q_6, \sqcup \rightarrow q_{\text{reject}}, \sqcup, R$
I_8	$q_2, 0 \rightarrow q_{\text{reject}}, \sqcup, R$	I_{20}	$q_6, 0 \rightarrow q_7, \sqcup, L$
I_9	$q_2, 1 \rightarrow q_3, \sqcup, L$	I_{21}	$q_6, 1 \rightarrow q_{\text{reject}}, \sqcup, R$
I_{10}	$q_3, \sqcup \rightarrow q_{\text{reject}}, \sqcup, R$	I_{22}	$q_7, \sqcup \rightarrow q_{\text{init}}, \sqcup, R$
I_{11}	$q_3, 0 \rightarrow q_{\text{reject}}, \sqcup, R$	I_{23}	$q_7, 0 \rightarrow q_7, 0, L$
I_{12}	$q_3, 1 \rightarrow q_7, \sqcup, L$	I_{24}	$q_7, 1 \rightarrow q_7, 1, L$

Table 1: [Transition Table of \$M\$](#)